

# PRAVEEN KUMAR MAHAWAR

PRINCIPAL AI SYSTEMS ARCHITECT | AUTONOMOUS AUTOMATION & FULL-STACK ENGINEER

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Full UK (GMT/BST) & US (EST/PST) Overlap

## EXECUTIVE PROFILE SUMMARY

High-velocity **Principal Systems Architect with 10+ years of deep engineering experience** architecting autonomous AI agent platforms, distributed web architectures, mission-critical analytics platforms, and high-scale headless automation pipelines. Founder and Lead Architect of **GARUDA OS** — an enterprise autonomous system coordinating multi-agent cognitive loops, stealth browser automation fleets, deterministic fintech state machines, and real-time electoral intelligence platforms processing 100,000+ data records with sub-100ms response times. Renowned for taking complex 0-to-1 systems into production without technical debt, enforcing strict zero-custody cryptographic guarantees, and executing with uncompromised velocity.

## CORE TECHNICAL ARSENAL

|                          |                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Autonomous AI & Agents:  | Multi-Agent Orchestration, Agentic Loops, Self-Healing Runtimes, Tool-Calling Pipelines, Persistent Synaptic Memory (JSONL/Vector), Context Compression. |
| Headless Automation:     | Puppeteer, Playwright, Chrome DevTools Protocol (CDP), Anti-Bot & Fingerprint Bypass, Proxy Rotation Farms, Distributed Web Scrapers.                    |
| Backend & Microservices: | Node.js (ESNext/CJS), Express, Python, RESTful API Engineering, WebSockets, Serverless Microservices (Vercel, Render), OAuth2/JWT Security.              |
| Frontend Architecture:   | React 19, Vite, Tailwind CSS, Dynamic Canvas/SVG Rendering, Offline Progressive Web Apps (PWA), Strict Viewport Anti-Flicker Architecture.               |
| Data & Cryptography:     | PostgreSQL, Supabase, Redis, SQLite, Deterministic State Machines, SHA-256 Ledger Auditing, Zero-Custody Enforcers.                                      |
| DevOps & Governance:     | Git Worktrees, Automated Build Verification Gates, Forensic Pre-Commit Linters, Cloudflare Edge, Linux & Windows Terminal Architecture.                  |

## PROFESSIONAL EXPERIENCE

Founder & Principal Systems Architect | GARUDA OS 2023 – PRESENT | REMOTE

- Autonomous Multi-Agent Architecture:** Engineered the core runtime of GARUDA OS — an enterprise autonomous system coordinating multi-agent task planning, self-healing code execution, and autonomous browser-based research workflows.
- Constituency War Room Platform:** Built an enterprise electoral intelligence platform ingesting 100,000+ official ECI Form 20 records across 12 battleground constituencies, delivering dynamic multi-quadrant voter polarity algorithms and real-time field cadre telemetry at 60 FPS.
- Headless Scraping & Signal Hunting Fleets:** Developed autonomous browser daemons executing authenticated session management, resilient DOM parsing, and target enrichment across high-security platforms with 99.4% uptime.
- Zero-Custody Fintech Infrastructure:** Implemented zero-custody financial qualification engines and multi-ledger reconciliation systems backed by immutable SHA-256 cryptographic hashing.
- 1-Shot Forensic Build Standards:** Enforced zero-regression governance across 450+ canonical routes, achieving clean exit-code 0 builds and sub-second asset streaming on production.

Lead Systems Consultant & Architect | High-Scale Automation Solutions 2018 – 2023 | REMOTE (UK / US / GLOBAL)

- High-Throughput Scraping Pipelines:** Built distributed data extraction infrastructure for international clients, harvesting and structuring over 500,000+ records daily from public registries, e-commerce catalogs, and dynamic web portals.
- API Integration & Middleware:** Built bidirectional connectors integrating Meta Graph API, Google Workspace, payment gateways, and custom webhook dispatchers with automated exponential backoff and dead-letter queues.
- Enterprise Web Platforms:** Delivered 20+ responsive web portals in React/Node.js for international businesses, reducing page load latencies by 55% via asset pre-rendering and edge caching.

- Developed robust MVC web applications, relational database schemas (MySQL/PostgreSQL), and customer portals for commercial enterprises.
- Automated manual reporting workflows into automated scheduled pipelines, saving clients 15+ hours of operational overhead weekly.

## FLAGSHIP ARCHITECTURAL SHOWCASES

**GARUDA Autonomous OS (garudaos.in)**  
Production platform running multi-agent cognitive layers, dynamic UI rendering, and pre-rendered SEO pipelines across 960+ static nodes.

**Constituency War Room & Cadre PWA**  
Mission-critical operational platform featuring dynamic QR matrices, offline-first booth telemetry, and multi-quadrant electoral math engines.

**CyberShield Threat Neutralization Engine**  
Perimeter defense framework detecting unauthorized brand pollution, security exposures, and DNS regressions with automated incident reporting.

**Stealth Browser Automation Fleet**  
Headless cluster bypassing anti-bot shields, handling 2FA tokens, and extracting multi-source signals without IP degradation.

## EDUCATION & CREDENTIALS

**Bachelor of Computer Applications (BCA)**

JABALPUR, MADHYA PRADESH, INDIA

Core Specialization: Systems Architecture, Database Management Systems (DBMS), Data Structures & Algorithms, Object-Oriented Programming, Operating Systems & Network Protocols.